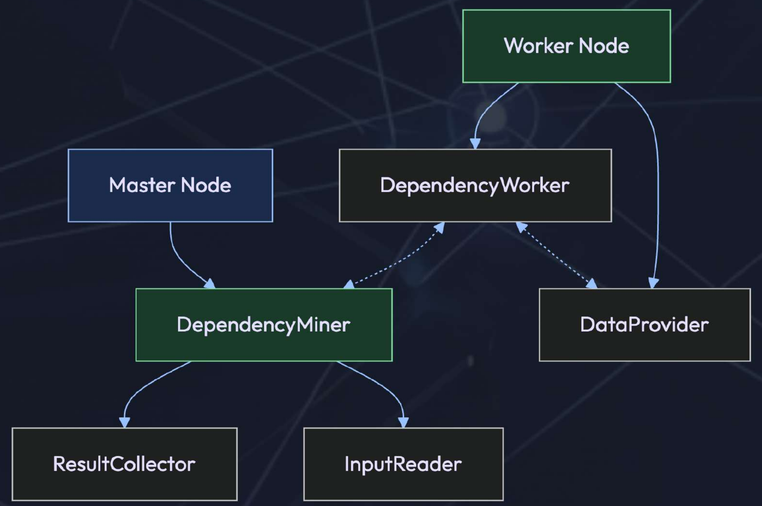
OVERVIEW

Actor Architecture

Architecture



System Strategy

- → **Dynamic Discovery** Akka Receptionist manages zero-config worker recruitment.
- → **Work-Pulling Pattern** Master assigns tasks only that are requested, preventing worker saturation.
- → **Local Data** DataProviders spread dataset columns for massive scalability.
- → **Smart Locality** Tasks are assigned to workers co-located with their data providers.
- → **Proxy Comms** LargeMessageProxy handles massive structure transfers seamlessly.

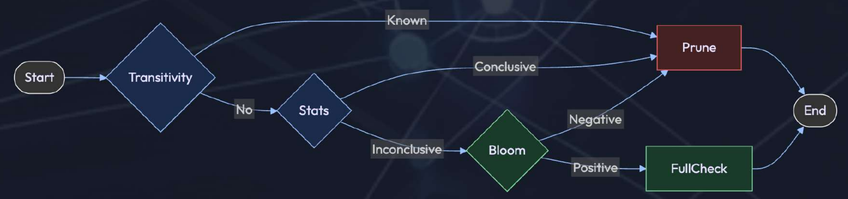
∞
SCALABLE NODES

95%
LOCALITY RATE

LOGIC & PERFORMANCE

Discovery Lifecycle

Pruning Flow



Optimizations

- → **O(n+m) Check** Sorted `ArrayList` ensures linear validation time.
- → **Custom Bloom** Optimized `BitSet` implementation with zero external dependencies.
- → **LHS Caching** Worker preserves recently used data to slash I/O overhead.
- → **Memory** Resource-safe monitoring prevents OOM on constrained nodes.

80-90%
PRUNING RATE

Linear
VALIDATION